

Computing Big-bang theory

Man's unending quest to understand the mysteries of the universe is now aided by Britain's biggest supercomputer based at the University of Durham.

This £1.4 million machine records data from billions of observations made on the behaviour of cosmic objects spread all across the universe. The data is then processed at ultra high speed to predict the formation and evolution of galaxies and solar systems. The analysis of the supercomputer is then compared to various cosmic theories to find a plausible explanation

for the evolution of the universe.

The supercomputer is being operated and maintained by the Institute of Computational Cosmology (ICC).

The supercomputer consists of an integrated cluster of 128 Ultra-SparcIII processors, 24 processor SunFire mid-frame servers, 112 Gigabytes of RAM and 7 terabytes of data storage, which allow it to perform 10 billion operations per seconds. This would take most human beings around a million years of continuous calculation to complete!

quicktake

Build a more secure environment with PKI technology

▲ What is PKI technology?

This is a technology, which involves building Public Key Infrastructure around the world. That means a network of trust that will spread across the world and will be used by organisations such as banks, governments and others. It includes digital certificates and validation. The digital certificates and PKI are governed through a protocol called OCSP, which

issues certificates on PA technology. Currently there are around 60 international banks using this technology to implement their businesses across the world. This technology is not proprietary and you can enjoy these services from other vendors as well such as Sun and IBM.

▲ What's your focus area in India?

In India we are part of Global Telecom Systems and are here to implement security solutions for Indian companies. We are currently focusing on all security aspects—from firewall to PKI. In matters of security, legal contracts, enforcement and cryptography all work together and it is we who are providing them. We cover business risks validation and security in data transfers.

When large volumes of data are transferred the network needs to be synchronised and we provide and maintain security for it. Our products work on a Java-based platform and is independent of the OS. It is customised and is served at the backend by an Oracle database.



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Sircam virus

To Sir with love

A W32 class virus called Sircam is creating havoc around the world by randomly sending personal messages from address books on infected machines. The rogue message is usually in the form of an e-mail attachment either in English or Spanish with a varying subject line. It could be "Hi! How are you? I send you this file in order to have your advice. See you later. Thanks" or "I hope you can help me with this file that I send" or somewhat similar. Your problems begin when you open the attachment. Opening it copies the worm to hidden files—SCAM32.EXE in the Windows/System directory and SIRC32.EXE—to the Recycled Bin folder. It also modifies a couple of registry keys.

Code Red worm

Red hot worm

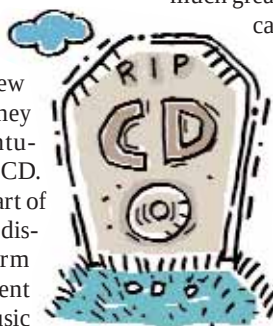
Experts claim that this worm has already caused more than \$1.2 billion worth of damage to

networks around the world. This worm was first recognised in mid-July and is programmed to infect computers specifically on the 1st and 28th of every month. Though it lies dormant for the rest of the period, it can cause considerable havoc with infected computers on a network with incorrect internal time and date settings. The Code Red worm takes advantage of a hole in Microsoft's Internet Information Server (IIS) Web server software and in Cisco servers.

There is speculation that the virus is the handiwork of the Chinese as some of the Web pages that were defaced had the line 'Hacked by Chinese'. The Netherlands-based Dutch group 29a has claimed authorship to a second variant of the Code Red virus. But the variable nature of the virus makes it hard to trace its origin. At last count there are at least two more variants doing the rounds—Code Red II and Code Red III—which could be even more harmful.

Bye, bye CDs

InterTrust, the company that specialises in Digital Rights Management (DRM) software, and DataPlay, a startup, are on the verge of releasing a new product, which they claim would eventually replace the CD. This would be a part of a portable media distribution platform for protected content such as music. Music distribution companies like Universal, EMI and BMG are all planning to release pre-recorded music on DataPlay digital media for use in multiple consumer electronic



devices. This new media would be able to hold about 11 hours of music, which is much greater than current CD capacities. The blank media would be priced at between \$5 and \$10 while pre-recorded ones will be priced more or less equivalent to the current CD price. The companies hope that markets would soon get used to the new media, and adopt it because of the greater capacity. Users would have to upgrade to devices that are compatible with the DataPlay media. →